Batch 1 - ML Lab Test Q1 - RF

Customer churn is a major issue for businesses, especially in subscription-based services. Accurately predicting whether a customer is likely to churn can help companies take proactive measures to retain them. Apply **Random Forest model** to analyse various customer features such as account tenure, total charges, monthly charges, contract type, and service usage to predict churn. Analyse the dataset and Predict behaviour to retain customers.

The data set includes information about:

- Customers who left within the last month – the column is called Churn.
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies.
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges.
- Demographic info about customers – gender, age range, and if they have partners and dependents.

1	Customer ID	Customer ID	Feature
2	**gender**	Whether the customer is a male or a female	Feature
3	**SeniorCitizen**	Whether the customer is a senior citizen or not (1, 0)	Feature
4	**Partner**	Whether the customer has a partner or not (Yes, No)	Feature
5	**Dependents**	Whether the customer has dependents or not (Yes, No)	Feature
6	**tenure**	Number of months the customer has stayed with the company	Feature
7	**PhoneService**	Whether the customer has a phone service or not (Yes, No)	Feature
8	**MultipleLines**	Whether the customer has multiple lines or not (Yes, No, No phone service)	Feature
10	**InternetService**	Customer's internet service provider (DSL, Fiber optic, No)	Feature
11	**OnlineSecurity**	Whether the customer has online security or not (Yes, No, No internet service)	Feature
12	**OnlineBackup**	Whether the customer has online backup or not (Yes, No, No internet service)	Feature
13	**DeviceProtection**	Whether the customer has device protection or not (Yes, No, No internet service)	Feature
14	**TechSupport**	Whether the customer has tech support or not (Yes, No, No internet service)	Feature
15	**StreamingTV**	Whether the customer has streaming TV or not (Yes, No, No internet service)	Feature
16	**StreamingMovies**	Whether the customer has streaming movies or not (Yes, No, No internet service)	Feature
17	**Contract**	The contract term of the customer (Month-to-month, One year, Two year)	Feature
18	**PaperlessBilling**	Whether the customer has paperless billing or not (Yes, No)	Feature
19	**PaymentMethod**	The customer's payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))	Feature
20	**MonthlyCharges**	The amount charged to the customer monthly	Feature
21	**TotalCharges**	The total amount charged to the customer	Feature
	Churn	Whether the customer churned or not (Yes or No)	Target

Using the Teleco Customer Churn dataset downloaded from Kaggle and made available in the local storage, implement Random Forest model to predict whether a customer will churn or continue using services based on features such as services, monthly and total charges. Evaluate the model's performance using metrics like accuracy, precision, and recall, and discuss the impact of feature selection on the results. (CO1, CO2, K5)

https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data

1) Load the dataset. (3 Marks)

2) Apply Exploratory Data Analysis and visualize the data. (2 Marks)

3) Drop irrelevant columns. (2 Marks)

4) Split dataset into features and target variable. (5 Marks)

5) Transform the categorical data into numerical. (3 Marks)

6) Apply Dimensionality Reduction technique (10 Marks)

7) Split the data into training and testing sets. (5 Marks)

8) Train the model on the prepared data. (5 Marks)

9) Predict the results. (5 Marks)

10) Experiment with different depth of the tree, number of trees and impurity metrics. (5 Marks)

11) Perform cross validation (3 Marks)

12) Evaluate the model using metrics such as accuracy, precision, recall, and F1-score.

Display the confusion matrix and analyze the results. (2 Marks)

Batch 2 - ML Lab Test Q2 - RF

Banks and financial institutions need to assess the risk associated with lending money to individuals. By analyzing a borrower's financial profile, credit score, loan history, and demographic factors, we can predict the likelihood of loan default. The goal is to develop a **Random Forest model** to classify whether a borrower will default on their loan, helping financial institutions make informed decisions. This dataset includes variables such as **loan amount, interest rate, debt-to-income ratio, credit score, and employment status**, making it ideal for classification using Random Forest.

	Column_name	Column_type	Data_type	Description
0	LoanID	Identifier	string	A unique identifier for each loan.
1	Age	Feature	integer	The age of the borrower.
2	Income	Feature	integer	The annual income of the borrower.
3	LoanAmount	Feature	integer	The amount of money being borrowed.
4	CreditScore	Feature	integer	The credit score of the borrower, indicating their creditworthiness.
5	MonthsEmployed	Feature	integer	The number of months the borrower has been employed.
6	NumCreditLines	Feature	integer	The number of credit lines the borrower has open.
7	InterestRate	Feature	float	The interest rate for the loan.
8	LoanTerm	Feature	integer	The term length of the loan in months.
9	DTIRatio	Feature	float	The Debt-to-Income ratio, indicating the borrower's debt compared to their income.
10	Education	Feature	string	The highest level of education attained by the borrower (PhD, Master's, Bachelor's, High School).
11	EmploymentType	Feature	string	The type of employment status of the borrower (Full-time, Part-time, Self-employed, Unemployed).
12	MaritalStatus	Feature	string	The marital status of the borrower (Single, Married, Divorced).
13	HasMortgage	Feature	string	Whether the borrower has a mortgage (Yes or No).
14	HasDependents	Feature	string	Whether the borrower has dependents (Yes or No).
15	LoanPurpose	Feature	string	The purpose of the loan (Home, Auto, Education, Business, Other).
16	HasCoSigner	Feature	string	Whether the loan has a co-signer (Yes or No).
17	Default	Target	integer	The binary target variable indicating whether the loan defaulted (1) or not (0).

Using the Bank Loan Default Prediction Dataset downloaded from Kaggle and made available in the local storage, implement Random Forest model to predict whether a customer will default or not based on features given in the dataset. Evaluate the model's performance using metrics like accuracy, precision, and recall, and discuss the impact of feature selection on the results. (CO1, CO2, K5)

https://www.kaggle.com/datasets/nikhil1e9/loan-default/data

1) Load the dataset. (3 Marks)

2) Apply Exploratory Data Analysis and visualize the data. (2 Marks)

3) Drop irrelevant columns. (2 Marks)

4) Split dataset into features and target variable. (5 Marks)

5) Transform the categorical data into numerical. (3 Marks)

6) Apply Dimensionality Reduction technique (10 Marks)

7) Split the data into training and testing sets. (5 Marks)

8) Train the model on the prepared data. (5 Marks)

9) Predict the results. (5 Marks)

10) Experiment with different depth of the tree, number of trees and impurity metrics. (5 Marks)

11) Perform cross validation (3 Marks)

12) Evaluate the model using metrics such as accuracy, precision, recall, and F1-score.

Display the confusion matrix and analyze the results. (2 Marks)

Batch 1 - ML Lab Test Q1- NB

Employee attrition is a major concern for businesses, as high turnover can lead to increased costs and loss of talent. By analyzing factors such as **job satisfaction, work experience, salary, number of promotions, working hours, and department**, develop a **Naïve Bayes classification model** to predict whether an employee is likely to stay or leave the company.

This model can help HR teams identify employees at risk of leaving and implement proactive retention strategies, such as improving job satisfaction and offering better career growth opportunities. Naïve Bayes is a suitable choice for this problem as it efficiently handles categorical and numerical features related to employee data.

This is a fictional data set created by IBM data scientists.

List of Features

Age
Attrition – Target
BusinessTravel
DailyRate
Department
DistanceFromHome
Education
EducationField
EmployeeCount
EmployeeNumber
EnvironmentSatisfaction
Gender
HourlyRate
JobInvolvement
JobLevel
JobRole
JobSatisfaction
MaritalStatus
MonthlyIncome
MonthlyRate
NumCompaniesWorked
Over18 OverTime
PercentSalaryHike
PerformanceRating
RelationshipSatisfaction
StandardHours
StockOptionLevel
TotalWorkingYears
TrainingTimesLastYear
WorkLifeBalance
YearsAtCompany
YearsInCurrentRole
YearsSinceLastPromotion
YearsWithCurrManager

Feature / Attribute Information with encoding applied on certain features listed below.

Education
1 'Below College'
2 'College'
3 'Bachelor'
4 'Master'
5 'Doctor'

EnvironmentSatisfaction
1 'Low'
2 'Medium'
3 'High'
4 'Very High'

JobInvolvement
1 'Low'
2 'Medium'

3 'High'
4 'Very High'
JobSatisfaction
1 'Low'
2 'Medium'
3 'High'
4 'Very High'
PerformanceRating
1 'Low'
2 'Good'
3 'Excellent'
4 'Outstanding'
RelationshipSatisfaction
1 'Low'
2 'Medium'
3 'High'
4 'Very High'
WorkLifeBalance
1 'Bad'
2 'Good'
3 'Better'
4 'Best'

Using the IBM HR Analytics Employee Attrition & Performance Dataset downloaded from Kaggle and made available in the local storage, implement Naïve Bayes model to predict whether an employee is likely to stay or leave the company based on features given in the dataset. Evaluate the model's performance using metrics like accuracy, precision, and recall, and discuss the impact of feature selection on the results. (CO1, CO2, K5)
https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
1) Load the dataset. (3 Marks)
2) Apply Exploratory Data Analysis and visualize the data. (2 Marks)
3) Drop irrelevant columns. (2 Marks)
4) Split dataset into features and target variable. (5 Marks)
5) Transform the categorical data into numerical. (3 Marks)
6) Apply Dimensionality Reduction technique (10 Marks)
7) Split the data into training and testing sets. (5 Marks)
8) Train the model on the prepared data. (5 Marks)
9) Predict the results. (5 Marks)
10) Experiment with different subsets of data. (5 Marks)
11) Perform cross validation (3 Marks)
12) Evaluate the model using metrics such as accuracy, precision, recall, and F1-score.
Display the confusion matrix and analyze the results. (2 Marks)

Batch 2 - ML Lab Test Q2 – NB

Accurate disease diagnosis is essential for early treatment and better healthcare outcomes. Many diseases share overlapping symptoms, making manual diagnosis challenging. The goal is to develop a **Naïve Bayes classification model** that predicts whether a patient has a particular disease based on input features such as **age, gender, medical history, vital signs, and reported symptoms**.

By leveraging the probabilistic nature of **Naïve Bayes**, we can model the relationships between different symptoms and diseases, providing an interpretable and efficient diagnostic tool. This model can assist healthcare professionals in preliminary diagnosis and decision-making, reducing misdiagnosis and improving patient care.

Using the Early-Stage Diabetes Risk Prediction Dataset downloaded from Kaggle and made available in the local storage, implement Naïve Bayes model to predict the risk of early-stage disbetes based on features given in the dataset. Evaluate the model's performance using metrics like accuracy, precision, and recall, and discuss the impact of feature selection on the results. (CO1, CO2, K5)

https://www.kaggle.com/datasets/ishandutta/early-stage-diabetes-risk-prediction-dataset

Feature / Attribute Information

Age 1.20-65
Sex 1. Male, 2.Female
Polyuria 1.Yes, 2.No.
Polydipsia 1.Yes, 2.No.
sudden weight loss 1.Yes, 2.No.
weakness 1.Yes, 2.No.
Polyphagia 1.Yes, 2.No.
Genital thrush 1.Yes, 2.No.
visual blurring 1.Yes, 2.No.
Itching 1.Yes, 2.No.
Irritability 1.Yes, 2.No.
delayed healing 1.Yes, 2.No.
partial paresis 1.Yes, 2.No.
muscle sti
ness 1.Yes, 2.No.
Alopecia 1.Yes, 2.No.
Obesity 1.Yes, 2.No.
Class 1.Positive, 2.Negative.

1) Load the dataset. (3 Marks)
2) Apply Exploratory Data Analysis and visualize the data. (2 Marks)
3) Drop irrelevant columns. (2 Marks)
4) Split dataset into features and target variable. (5 Marks)
5) Transform the categorical data into numerical. (3 Marks)
6) Apply Dimensionality Reduction technique (10 Marks)
7) Split the data into training and testing sets. (5 Marks)
8) Train the model on the prepared data. (5 Marks)
9) Predict the results. (5 Marks)
10) Experiment with different subsets of data. (5 Marks)
11) Perform cross validation (3 Marks)
12) Evaluate the model using metrics such as accuracy, precision, recall, and F1-score.
Display the confusion matrix and analyze the results. (2 Marks)